body3d: An Anatomical Muscle-Activation Viewer for the Web, Shaped by Fat-Free Mass and Tape Measures

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Abstract

Fitness and rehabilitation applications tell people which muscles an exercise works, usually with flat diagrams of a generic body. We present **body3d**, an open-source viewer that shows muscle activation on an anatomical 3D body in a web page, and makes that body resemble the person. An offline pipeline turns the BodyParts3D database into a rigged model of 135 muscles per side over a full skeleton. A layered hypertrophy model grows the muscles without interpenetration; because it is linear in the growth emphasis, splitting it by body region yields exactly additive morph targets calibrated against the fat-free mass index (FFMI). A regularized least-squares fit then adjusts the regions to a person's tape measures while conserving the muscle volume implied by their FFMI, with a population prior learnt from the ANSUR II anthropometric survey. On held-out people it predicts a hidden limb girth with an RMS error of 1.5–1.6 cm, against 3.3–3.7 cm for the population mean: as accurate as a ridge regression given only height, weight and body fat, but 0.2–0.3 cm behind it when every other tape is given — the measured cost of explaining all tapes with one anatomical body. Body composition in this evaluation is itself estimated from tapes. The viewer also chooses a default camera for any activation set. Everything runs in three.js without decoders or WebAssembly, at 2.3 ms per frame on a desktop GPU and about 10 ms per fit.

1 Introduction

"Which muscles does this exercise work?" is among the most common questions in strength training, and applications answer it with a highlighted silhouette. Two shortcomings of that answer motivated this work. First, a 2D map hides depth and side: the triceps is behind the arm, the gluteus medius sits above and lateral to the maximus, the brachialis lies under the biceps. Second, the body shown is nobody in particular, while the same applications store the person's weight, body fat and girths.

body3d is a web component that addresses both: an anatomical body of individually named muscles over a skeleton, coloured by activation, that can be orbited, tapped and posed, and whose shape follows the person's fat-free mass and tape measures. It must run in ordinary web pages — on phones, inside applications with strict Content-Security-Policies that forbid WebAssembly and `eval` — and must be embeddable as a library, a custom element or an iframe.

Our contributions are:

- a reproducible pipeline from the BodyParts3D anatomical database to a rigged, quantized glTF model, including the synthesis of a missing muscle by elastic fibres draped over a distance field (Section 4);
- a layered hypertrophy model whose region-wise split is *exactly additive*, giving 14 morph targets calibrated against fat-free mass (Section 5);
- a tape-measure fit that reconciles regional girths with total muscle mass, with closed-form girth and volume models, a population prior calibrated on ANSUR II and robust outlier handling, evaluated out of sample against statistical baselines (Sections 8 and 9);
- an automatic choice of the camera for an activation set, from per-group visibility in a picking buffer (Section 7);
- a runtime design for the web: per-part state in a data texture, dual-quaternion skinning on quantized geometry, GPU picking and on-demand rendering (Section 6).

We also report what the evaluation could not settle — real left–right asymmetry, a trained population, agreement with coaches on views — and release the protocols and scripts for those studies.

2 Related work

Anatomical data. BodyParts3D [27] maps concepts of the Foundational Model of Anatomy [29] to polygon meshes of one adult male body; since 2025 it is distributed under CC BY 4.0. It is, to our knowledge, the only open dataset with volumetric, individually named muscles tied to an ontology. Statistical body models such as SMPL [23] describe the skin surface only; musculoskeletal simulators such as OpenSim [4] represent muscles as lines of action.

Anatomical body modelling. Graphics research has built detailed, simulated musculature: Teran et al. [34] extracted and simulated muscles from the Visible Human data; Lee et al. [21] modelled and controlled the upper body with hundreds of muscles; Saito et

al. [30] grew muscle and fat with physics-based growth and elasticity to produce a range of body shapes from one anatomical template; Kadleček et al. [14] fitted a template anatomy to 3D surface scans. These methods solve volumetric elasticity problems offline or near-interactively and take scans as input.

The inside from the outside. A line of learned models infers internal anatomy from the body surface of SMPL. OSSO [16] predicts the skeleton from the skin, trained on paired DXA scans; SKEL [17] re-rigs SMPL with a biomechanically accurate skeleton. HIT [18], the closest to our goal, learns from segmented MRI registered to SMPL an implicit model of where subcutaneous fat, lean tissue (muscles and organs together) and long bones lie inside a body of given shape and pose. body3d differs in what it models and from what: named individual muscles rather than tissue classes, inferred from the tape measures and body composition people record rather than a surface scan, by a small calibrated optimization rather than a learned network, and evaluated by a GPU in a web page.

Bodies from measurements. Synthesizing a body from anthropometric measurements is an old problem for skin surfaces. Seo and Magnenat-Thalmann [32] learnt from range scans an interpolation from sizing parameters to body shape; Wuhrer and Shu [37] estimated shapes from CAESAR-type measurements with a statistical shape model and a nonlinear optimization that extrapolates beyond the training set; Anny [3], an open parametric model, spans ages and body types through semantic phenotype parameters calibrated on WHO statistics. body3d solves the analogous problem one layer down: the musculature behind given girths, with a prior on muscle per region.

Body composition and anthropometry. The fat-free mass index (FFMI) normalized by height [19, 31] measures muscularity independently of fat; skeletal muscle mass relates to fat-free mass through whole-body MRI studies [1, 13, 26], and corrected girths (girth minus π times the skinfold) [9, 20] relate tapes to muscle cross-sections. The US Navy method [10] estimates body fat from girths. ANSUR II [8] provides 93 measurements on 6 068 U.S. Army personnel.

3 System overview

Everything expensive happens offline (Fig. 1): mesh selection and cleaning, synthesis, joint placement, skinning weights, hypertrophy morphs, baked ambient occlusion and the data of the tape fit. The browser receives a self-describing glTF file [35] per model (male and female, each at two resolutions: 380 k and 159 k triangles) and performs only what reacts to the person. The calibration is computed by a separate step on the exported fit data; because it depends on the exact geometry, it is keyed by a hash of those data, and a rebuilt model without a matching calibration fails the test suite.

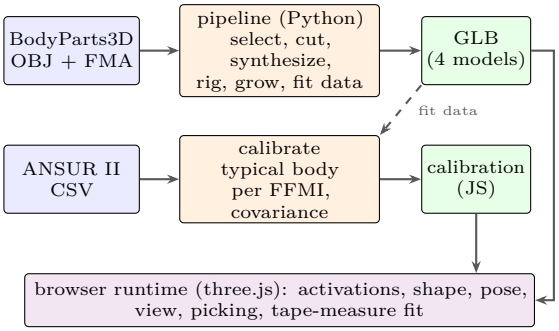


Figure 1: Offline pipeline and runtime. The calibration is computed on the exported model's fit data and keyed by its fingerprint.

The pipeline is deterministic: a rebuild from the same sources produces byte-identical geometry.

4 Building the anatomical model

Selection. Of the 2 234 BodyParts3D meshes (426 muscles, 225 bones, 30 cartilages), an explicit table lists 133 muscles per side by concept name, three connective tissues, rules assigning bones to 52 rig segments, and the viscera used later as a hypertrophy core; with the two structures added below — the synthesized latissimus dorsi and the rectus abdominis cut out of the abdominal wall — the model has 135 muscles per side. Table-driven selection replaced name-based heuristics after two failures: a pattern meant for foot bones matched a laryngeal cartilage, and one file named "left" lies in the right hand. The build stops on any inconsistency (missing names, a part with pieces on both sides, a bone far from its segment).

Vocabularies. Muscles belong to 44 anatomical groups (e.g. the three parts of the pectoralis and of the deltoid, gastrocnemius separate from soleus), labelled in English and Brazilian Portuguese. A 25-group training vocabulary ("Chest", "Lats", "Upper Back") maps onto groups with weights, e.g. "Brachialis" activates the brachialis at 1 and the brachioradialis at 0.5. Names resolve case- and accent-insensitively with a fixed precedence.

Cut and synthesized structures. In BodyParts3D the external-oblique mesh is the whole anterior abdominal wall. We cut the rectus abdominis out along the linea semilunaris, measured as a groove on horizontal slices (5 cm from the midline near the pubis, 6.3–7 cm at the navel, 9 cm at the costal margin), by splitting triangles on the zero set of a scalar field so both parts stay glued.

The database has no latissimus dorsi. We synthesize it as $K = 36$ fibres from a fleshy origin measured on this body (spinous processes under the trapezius, the lateral border of the erectors, the iliac crest) to the intertubercular sulcus of the humerus. Let D be the signed distance to a *support* — ribs, spine, scapula, pelvis, deep muscles and viscera, closed morphologically

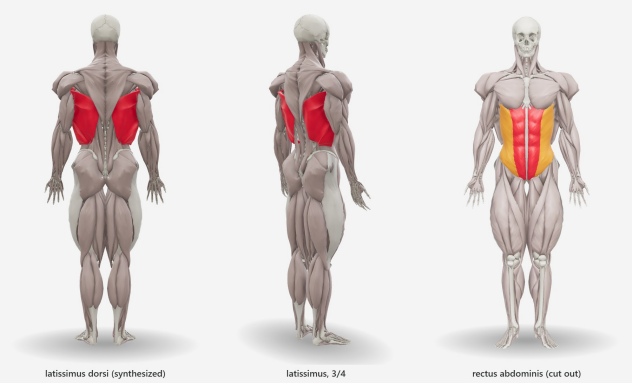


Figure 2: The synthesized latissimus dorsi (left, centre) and the rectus abdominis cut out of the abdominal wall (right).

and filled slice by slice so no fibre crosses the thorax. Each fibre $\{\mathbf{p}_i\}$ starts as an arc around the trunk and alternates a shortening step and a projection:

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \tfrac{1}{2}\big(\tfrac{1}{2}(\mathbf{p}_{i-1} + \mathbf{p}_{i+1}) - \mathbf{p}_i\big), \tag{1}$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \max\big(0, \delta - D(\mathbf{p}_i)\big)\,\nabla D/\|\nabla D\|, \tag{2}$$

until it lies taut over the support (multiresolution: 9, 17, 33, 65, 96 points). The sheet of fibres receives a thickness by volume conservation — proportional to the width at the origin over the local width, capped at 2 cm — tapering at the free borders and into a flat tendon (Fig. 2). The thoracolumbar aponeurosis is deliberately not drawn: it would hide the erectors, which are activated in many exercises.

Geometry. Meshes are welded, decimated by quadric error [7] to a budget spread by surface area, and smoothed by Taubin's $\lambda|\mu$ filter [33] with fixed borders, which does not shrink and is linear. Volumetric operations use a voxelization by ray parity along three axes with a majority vote, robust to the database's non-watertight meshes (a flood fill inflated the left thigh, whose femur leaks on the grid).

Rig and weights. Fifty-two joints have identity rotation in the rest pose and are placed from bone geometry: least-squares spheres on the femoral and humeral heads, the midpoints of epicondyles and malleoli, the centres of intervertebral disks, and bone contacts for wrists and fingers. Skinning weights are inverse cubed distances to bone surfaces, restricted to the segments that dominate a part and their chain neighbours, with the dominant segment weighted 4×; without these rules a hand hanging by the thigh followed the leg, and the biceps stretched across the elbow.

Female proportions. BodyParts3D has one male body. The female model applies per-segment affine transforms through the skinning weights (pelvis width ×1.14, thorax ×0.92, clavicles ×0.88, +4° carrying angle, knee valgus, smaller hands, feet and neck) before the hypertrophy, followed by a female distribution of muscle gain.

5 Layered hypertrophy

From FFMI to muscle volume. Body size is expressed by the normalized FFMI [19],

$$F = \frac{\mathrm{FFM}}{h^2} + 6.1\,(1.8 - h), \tag{3}$$

with fat-free mass FFM in kg and height h in metres. Skeletal muscle mass is $\mathrm{SMM} = k(F)\,\mathrm{FFM}$. Whole-body MRI puts SMM at 38 % and 31 % of body mass in men and women [13], which with the body fat of those samples gives $k \approx 0.49$ and 0.47. The ratio is higher in trained people — 0.477 against 0.447 in college athletes and non-athletes [26], about 8 % higher in very large athletes [1] — so k rises linearly to 0.56 at $F = 26$ for men (0.52 at 22 for women). The direction is documented; the slope, about twice the reported differences, is our assumption. Its reach is limited: after the calibration of Section 8 it only sets the trunk of the typical body, and half the slope would make the default male's shoulders 2.7 cm narrower. The represented muscle volume is $V = c\,\mathrm{SMM}/\rho$ with $\rho = 1.06$ kg/L and a coverage $c = 0.92$ for the deep muscles the model lacks. The base body is 1.707 m tall with 18.7 L of muscle, $F \approx 15$; at the default $F = 26$ the male model carries 39.6 L.

The stacking integral. Inflating each muscle along its normals makes layers cross. Instead, let $\tilde{D}$ be the smoothed distance to the *core* (bones, cartilage and viscera), $\mathbf{u} = \nabla\tilde{D}/\|\nabla\tilde{D}\|$ and $e(\mathbf{x})$ the growth emphasis of the muscle at $\mathbf{x}$ (zero outside muscle). A surface point moves by

$$\mathbf{d}(\mathbf{p}) = g\,\mathbf{u}(\mathbf{p}) \int_0^{\ell(\mathbf{p})} e\big(\mathbf{p} - s\,\mathbf{u}(\mathbf{p})\big)\,ds, \tag{4}$$

the thickening of all muscle between the point and the core, with ℓ bounded by 1.3 times the true distance to the core. The inner face of a muscle moves with the outer face of the one below, so layers remain stacked by construction; in the abdomen the core is the viscera, so the wall thickens without the belly inflating. The global gain g is solved so that the muscle volume meets the target; the emphasis (e.g. lateral deltoid 1.65, obliques 0.6 for men; gluteus maximus 1.6 for women) distributes the gain like a trained physique.

Exactly additive regional morphs. Partition the muscle groups into $R = 14$ regions (upper arm, forearm, thigh and lower leg per side; shoulders, chest, back, neck, core, hips) and write $e = \sum_r e_r$. Eq. (4) is linear in e, and every later step — the isotropic limbs below, vector smoothing over the mesh, Taubin smoothing with fixed borders, the uniform scale to the target height — is linear in the displacements. Therefore the per-region displacements $\mathbf{m}_r$ obtained from e_r satisfy, exactly,

$$\mathbf{x}(\mathbf{w}) = \mathbf{x}_0 + \sum_{r=1}^{R} w_r\,\mathbf{m}_r, \tag{5}$$

and $w_r = t$ for all r reproduces the uniform body at gain t. The pipeline checks the sum on every vertex

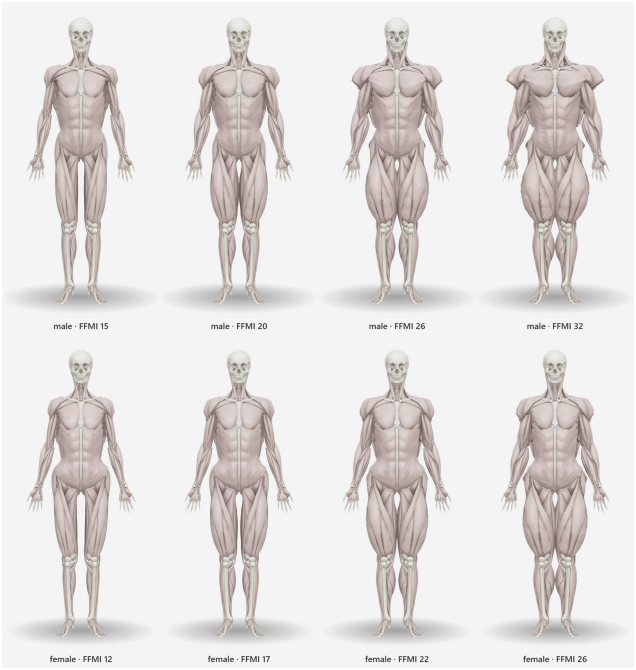


Figure 3: The male model at FFMI 15, 20, 26 and 32 and the female model at 12, 17, 22 and 26 (typical bodies, Section 8).

(the largest difference is 1.5×10^{-15} m, float64 rounding) and exports $\mathbf{x}_0$ at the minimum of the FFMI range and the $\mathbf{m}_r$ as glTF morph targets, with a table from F to the uniform weight (Fig. 3).

Isotropic limbs. Stacking thickens a layer along the thickness it already has; the arm, whose biceps and triceps are thick antero-posteriorly, grew almost only in depth (a 10×21 cm section at $F = 26$). Mid-limb, we replace the displacement by a radial scale about each section's centre, blended in by limb and faded near the joints. The scale of a level, $s = \sqrt{1 + P\bar{m}/A}$ for a hull of area A, perimeter P and mean outward displacement $\bar{m}$, is computed *once*, from the stacking displacement per unit gain on the rest shape, and enters as a fixed field: the hypertrophy stays linear and the morphs exactly additive. It matches the outline area that stacking adds at unit gain only to first order, and does not conserve each level's volume. Measured against plain stacking over the drawable range, a limb's own muscle volume changes by -9 to $+23$ % and its mid-limb outline area by -14 to $+18$ %, and the upper arm's field enlarges the latissimus near the axilla, which stacking would only have translated. The FFMI table is computed after this step, on the measured volume, so the drawn FFMI stays calibrated; over the whole body the limbs' share of the uniform growth moves from 68.7 to 69.4 % (male).

6 Runtime

Format. Positions are int16 and normals int8 (KHR_mesh_quantization), skin weights and ambient occlusion uint8, indices uint16; three.js [36] reads them natively, so no decoder, worker or WebAssembly is needed. One quantization box covers the body and its dequantization is folded into the inverse bind matrices, so a single skin serves all meshes. Soft tissue is split into 14 meshes by dominant region, each carrying only the 1–10 morph targets that move it; three quarters of the targets are sparse. The full models are 11 MB (5.7 MB gzipped), the lite ones 5 MB (2.6 MB).

Shading. All per-part state (activation, highlight, dimming, kind) lives in a 1×547 RGBA data texture indexed by a per-vertex part attribute: changing what is active rewrites at most 2.2 kB, and the body remains 15 draw calls. The material is three.js's physically based material with replaced shader chunks: an activation colour ramp, a rim term, baked ambient occlusion, and *dual-quaternion skinning* [15], which preserves volume under the twists of the bodybuilding poses where linear blending collapses shoulders and elbows. Because the bone matrices carry the uniform dequantization scale, the shader extracts it from the first column before converting the rotation to a quaternion and reapplies it before the rigid part.

Picking and rendering. A tap renders a 1×1 off-screen pass in which every fragment writes its part index; the answer is exact on the skinned, morphed, posed body at the cost of one pixel (1.1 ms on a desktop GPU), without a CPU copy of deformed geometry. The render loop draws only after a change. Six classic poses per sex are rotation lists about body axes, grounded by lowering the body until the lowest foot sample touches the floor (Fig. 4).

7 Choosing a view

A view is stored relative to the body as yaw, pitch, zoom and a focus point in body heights, so it fits either model, any FFMI and any viewport. For an activation set a_i on parts i, let groups $\mathcal{G}$ have weights $\omega_g = \max_{i \in g} a_i^2$. We render the picking pass at 90×150 pixels from eight azimuths (front, back, sides, the four 3/4 views) at pitches 4.6° and 16°, count the pixels $n_g(v)$ of each group in each view v, and score

$$S(v) = \beta(v) \sum_{g \in \mathcal{G}} \omega_g \, \frac{n_g(v)}{\max_{v'} n_g(v')}, \tag{6}$$

with $\beta = 1.10$ for the front, 1.07 for the back and 0.94 for the raised camera. Normalizing per group makes a group count by how well a view shows it relative to its best view, not by its size: with raw pixel counts the large trapezius out-voted the lateral deltoid and lateral raises were shown from behind. Squaring the activation lets secondary muscles break ties only. In the winning view, the bounding box of the key muscles (weight $\geq \frac{1}{2} \max \omega$) sets a zoom of $1/(1.6 \max(b_w, b_h))$, capped at 1.5, and a focus at its centre, unless the box spans more than 62 % of the height. The computation takes 23 ms on a desktop GPU.

Applied to the 372 exercises of a training catalog (primary muscles at 100 %, secondary at 45 %), the method chose the back for 202, the front for 138, a front



Figure 4: Six poses per model, each with the activations it suggests, rendered with dual-quaternion skinning.

3/4 view for 27, a back 3/4 view for 3 and a side view for 2 (decline presses), zooming in for 357 (Fig. 5). These views were reviewed visually by the author only; a study with coaches is prepared (Section 10). Views are stable but not invariant across model versions: computed on the body before the ANSUR II calibration, 345 of the 372 had the same azimuth, most others flipping between the two mirror-image front 3/4 views, which score almost equally on a nearly symmetric body. When key muscles lie on opposite faces, a *split view* renders the diametrically opposite camera in the other half of the canvas, synchronized so both halves always show opposite sides.

8 Fitting the body to tape measures

Observation model. For each of 12 tape sites (arm flexed, forearm, thigh, calf on both sides; neck, shoulders, chest, hips), the pipeline stores cross-sections perpendicular to the limb axis — or horizontal on the trunk — at several levels. A tape bridges concavities, so the girth is the perimeter of the convex hull of the section, at the thickest level for limbs and the thinnest for the neck. Re-slicing the mesh in the browser is too slow; instead each section stores *material points*, the crossings of mesh edges with the plane in a reference shape, with their per-region displacements: $\mathbf{q}_j(\mathbf{w}) = \mathbf{b}_j + \sum_r w_r \mathbf{d}_{rj}$. The girth $G_s(\mathbf{w})$ is the hull perimeter (Andrew's monotone chain [2]); since the hull's vertex set is locally constant, its gradient is analytic,

$$\frac{\partial G_s}{\partial w_r} = \sum_{(i,j)\in \mathrm{hull}} \frac{(\mathbf{q}_j - \mathbf{q}_i)^\top}{\|\mathbf{q}_j - \mathbf{q}_i\|} (\mathbf{d}_{rj} - \mathbf{d}_{ri}). \tag{7}$$

The signed volume of a closed mesh, $\frac{1}{6}\sum_t \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$, is with affine vertices an *exact cubic polynomial* in $\mathbf{w}$; the GLB stores its ≈ 500 monomials.

A tape wraps skin and subcutaneous fat. For a ring of thickness t around a convex section the perimeter grows by $2\pi t$ (Steiner), so with skinfold $S \approx 2t$ the muscle-and-bone girth is $C - \pi S$ [9, 20]. Without skinfolds, the skinfold of a site is estimated from total fat mass FM and body surface area [28]:

$$S = 2.5\,\mathrm{mm} + a(\mathrm{site}, \mathrm{sex})\,(\mathrm{FM}/\mathrm{BSA})^{0.9}, \tag{8}$$

with coefficients reproducing the mean skinfolds of Lee et al. [20] within 3 mm and a 20 % individual variation entering each measure's uncertainty — independently per site, a simplification the evaluation shows to be costly. Girths are carried to the model's height through the ratio of lean masses at the normalized FFMI, divided by the height ratio (areas), square-rooted (perimeters).

Objective. The unknowns are the region weights $\mathbf{w}$ and the FFMI F. We minimize

$$E(\mathbf{w}, F) = \sum_s \rho_H\Big(\frac{G_s(\mathbf{w}) - y_s}{\sigma_s}\Big) + \tfrac{1}{2}\,\boldsymbol{\Delta}^\top \Sigma^{-1} \boldsymbol{\Delta} + \tfrac{1}{2}\Big(\frac{V(\mathbf{w}) - V^\star(F)}{0.01\,V^\star(F)}\Big)^2 + \tfrac{1}{2}\Big(\frac{F - F_0}{\sigma_F}\Big)^2, \tag{9}$$

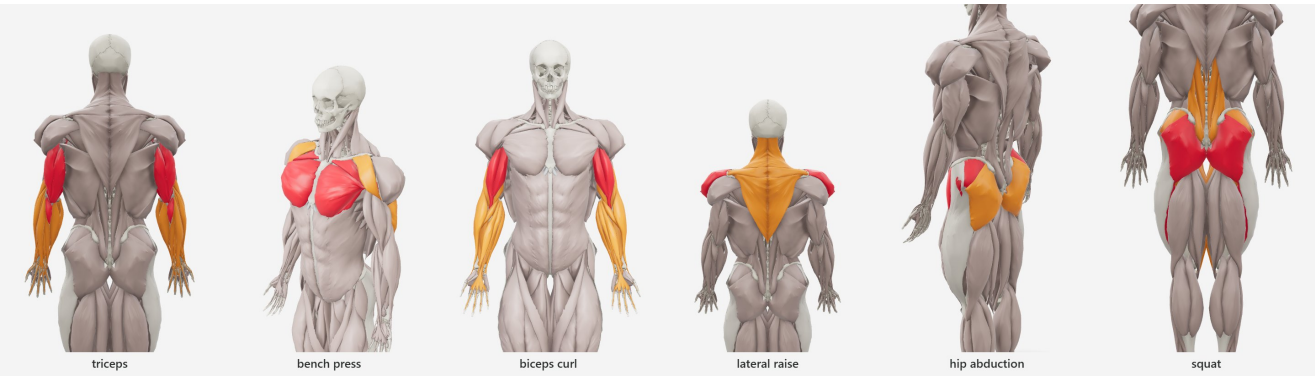


Figure 5: Automatic views. The squat (right) drives quadriceps and glutes equally from opposite faces; any single view hides half, which the split view addresses.

Table 1: RMS error (cm) predicting a hidden tape on 600 unseen people per sex. Mean: the calibration half's sex mean. Then three information sets — height, weight and body fat (h,w,f); plus one neighbouring tape (+1); plus every other tape (all) — each predicted by a ridge regression trained on the calibration half (R) and by the fit (F). †The Navy body fat reads the neck (and the hips for women), so the h,w,f columns are not blind for those sites.

	Men							Women						
		h,w,f		+1		all			h,w,f		+1		all	
Site	mean	R	F	R	F	R	F	mean	R	F	R	F	R	F
Biceps	3.45	1.65	1.60	1.33	1.36	1.26	1.51	2.88	1.33	1.42	1.17	1.28	1.18	1.33
Forearm	2.16	1.10	1.09	0.89	0.92	0.86	0.87	1.75	0.94	0.93	0.83	0.88	0.79	0.84
Thigh	5.96	2.30	2.60	2.26	2.54	1.59	2.46	5.61	2.21	2.52	2.13	2.43	1.76	2.33
Calf	3.06	1.66	1.66	1.64	1.61	1.55	1.58	2.88	1.65	1.67	1.59	1.66	1.46	1.53
Limb mean	*3.66*	*1.68*	*1.74*	*1.53*	*1.61*	*1.31*	*1.61*	*3.28*	*1.53*	*1.64*	*1.43*	*1.57*	*1.30*	*1.51*
Neck†	2.63	1.39	1.37	1.37	1.45	1.23	1.50	1.83	1.19	1.19	1.14	1.22	1.06	1.13
Shoulders	6.44	3.18	3.09	2.91	3.05	2.87	3.05	5.04	2.92	2.89	—	—	2.72	2.92
Chest	8.69	3.34	4.55	3.06	4.23	2.65	4.43				(not used)			
Hips	7.77	2.55	2.85	1.99	2.20	1.95	2.26	7.42	2.12†	2.62†	1.78	2.16	1.79	2.36

with $\boldsymbol{\Delta} = \mathbf{w} - \boldsymbol{\mu}(F)$, where y_s is the corrected, height-scaled tape; σ_s combines tape technique (1 %), level placement (1.2–4 %) and the fat estimate; ρ_H is the Huber loss [12] ($k = 2.5$); $\boldsymbol{\mu}(F)$ and Σ are the typical body of FFMI F and the between-region covariance; $V^\star(F)$ the muscle volume of the typical body; and F_0, σ_F the FFMI from weight, height and body fat with the uncertainty of the body-fat method (or the population mean with a wide deviation when no body fat is known; the US Navy formula [10] is used when girths allow it). The volume term is what makes a thigh larger than expected take muscle from the other regions when the FFMI is well measured, and raise the FFMI when it is not. Trunk girths also reflect skeletal breadth and fat distribution, which the model does not vary; they enter as a ratio over the population mean at the same FFMI, applied to the typical body's section. We solve with Levenberg–Marquardt [22, 25] with an active set at the weight bounds $[-0.15, 1.35]$, iteratively reweighting the Huber terms; it converges in 3–6 iterations. The inverse normal matrix at the optimum gives per-region uncertainties. A tape is flagged as an outlier and removed when it deviates by more than six *population* standard deviations from the prediction of a fit without it.

Calibration on ANSUR II. For each of the 4 082 men and 1 986 women of ANSUR II [8] we compute the Navy body fat, the normalized FFMI (men 21.7 ± 1.9, women 17.7 ± 1.0) and every tape in model space as the fit does. Per site we fit the cross-section area (girth^2) linearly in F; the typical body $\boldsymbol{\mu}(F)$ reproduces these curves on the limbs and keeps the model's hypertrophy on the trunk. ANSUR II measures the limbs on one side only, so both sides are tied, with an *assumed* between-side deviation of 2 % of the girth. The covariance of the deviations at equal FFMI, minus the modelled noise of tapes and estimated fat (at most 80 % of the observed variance) and projected to the nearest positive-definite matrix, is carried to region weights through the Jacobian. In the fit it is scaled by a factor of 3 in standard deviation, a design choice for training populations, which we expect to vary more than soldiers (Section 9); the correlations, which propagate information from measured to unmeasured regions, are kept. Calibration replaced appearance-tuned proportions: at $F = 26$ the arm had carried 51 cm of muscle alone, more than a natural bodybuilder's flexed tape.

9 Evaluation

Geometric fidelity. Sampling region weights across the plausible range, the probe girths match true cross-sections of the deformed meshes within 0.4 % (arms), 0.2 % (forearms, calves), 1.0 % (neck), 1.6 % (shoulders), 2.0 % (hips) and 2.9 % (thighs); the volume polynomial matches the mesh to 10^{-6} relative; the region morphs sum to the uniform morph to float rounding.

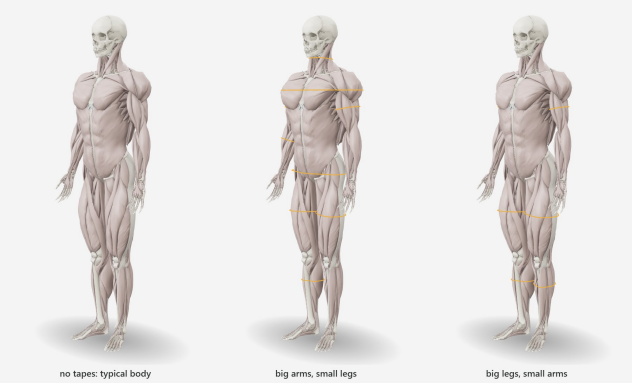


Figure 6: One person (178 cm, 84 kg, 14 % body fat). Left: no tapes, the typical body of the FFMI. Centre and right: the same FFMI with large arms or large legs; muscle moves between regions and the rings show the fitted tapes, including the estimated fat.

Table 2: Runtime cost (ms, medians). GPU: RTX 4060 Ti; CPU: SwiftShader software rasterization.

	GPU		CPU	
	full	lite	full	lite
Frame	2.3	1.3	256	156
Split-view frame	3.5	2.0	—	—
Pick	1.1	0.8	26	12
Best view	23	15	589	268
Tape fit (8 tapes)	6.5	9.2	—	—

Anatomical plausibility. The fit constrains hull perimeters and total volume, never the inside. As a coarse check we compare the right quadriceps of the typical body, computed from the exported meshes, with MRI volumes of 52 untrained and 16 long-term resistance-trained young men [24] (Table 3). That study reports no body composition, so we sweep a plausible body fat; a person's quadriceps is the typical body's at their FFMI, scaled by their lean mass. The group volume falls within −0.6 to +1.4 standard deviations of the measured means, but the model underestimates the difference between the groups (+37 to +39 % for the same body-fat gap, +57 % measured): trained men carry more quadriceps than the typical body of their FFMI. The split between heads disagrees, at every FFMI: BodyParts3D's rectus femoris is a quarter of the quadriceps and its vastus intermedius less than a fifth, against 13 % and 30 % by MRI. Volumes are plausible per muscle group — what the viewer colours — not per head.

Out-of-sample accuracy. ANSUR II has no reference body composition, so each person's body fat is estimated — by default with the US Navy formula on their own neck, waist and (women) hips. Body composition is therefore a proxy throughout, and the agreement between fitted and input FFMI below is not an independent check. We calibrated on the even-numbered rows of the public file (2 041 men, 993 women) and tested on the first 600 odd-numbered rows of each sex, with no further selection. Each tape, hidden in turn, is predicted from height, weight and body fat alone (the typical body), with one neighbouring tape, and with all the others, and compared with two baselines trained on the calibration half on exactly the same inputs: the sex mean and a ridge regression [11] with a cross-validated penalty (Table 1).

Table 3: Right quadriceps volume (cm^3): MRI of 52 untrained (176 cm, 73 kg) and 16 trained men (183 cm, 91 kg) [24], mean ± SD, and the model over an assumed body fat. Last row: share of each head (vastus lateralis, intermedius, medialis; rectus femoris).

	MRI	Model	Body fat
Untrained	1838 ± 263	1943–2216	20–12 %
Trained	2882 ± 308	2704–3041	16–8 %
Heads, %	33 / 30 / 24 / 13	37 / 18 / 20 / 25	

The typical body of an FFMI halves the error of the mean and is within 0.1 cm of the regression on the limbs given the same three numbers (limb mean 1.74 against 1.68 cm for men). With more tapes the regression pulls ahead: given every other tape it is better by 0.2–0.3 cm on the limbs, 0.9 cm on the men's thigh and 1.8 cm on the chest. Explaining every tape with one anatomical body costs accuracy once many tapes are given. A diagnostic run locates the cost: the regression's thigh error drops from 2.24 to 1.59 cm when the trunk tapes are added — a large hip girth predicts a large thigh through shared fat and pelvic breadth — while the fit, whose prior holds only the covariance of *muscle* between regions and whose per-site fat deviations are independent, barely uses them (2.52 to 2.46 cm). The same mechanism explains why more tapes sometimes hurt the fit (one neighbouring tape beats all of them for the men's biceps, neck, chest and hips): the men's biceps is predicted to 1.33 cm from the other limb tapes and to 1.51 cm once the trunk tapes are added. Trunk tapes enter as ratios and carry mostly fat and skeletal breadth; with a wide prior the fit moves trunk regions to match them and the muscle correlations drag the limbs along. A regression, which weighs each input by its predictive value, never loses from more inputs. Biases stay below 0.4 cm, no real body was flagged as an outlier, and the fitted FFMI stays within ±0.75 of the input for 90 % of people. Fits take 7–15 ms on average in Node.js (Intel i7-12700F), varying between runs.

With a girth-free body fat — Deurenberg's formula on BMI and age [5] — the typical-body column becomes blind to every tape: its limb error rises to 1.88 cm (men) and 1.69 cm (women), still half the mean's, and biases appear, because BMI attributes muscle to fat in muscular soldiers (the fit raises their FFMI by 0.34 on average to reconcile the tapes).

Asymmetry. ANSUR II measures one side, so real left–right differences cannot be validated on it. Imposing a synthetic difference instead (Table 4), the fitted body reproduces 80–85 % of it for men and 48–66 % for women; the rest is absorbed by the prior that ties the sides and, through the volume term, by the measured side, which moves by up to 0.35 cm. Women's tapes

Table 4: Share of a synthetic left–right difference (left tape = right $\times(1+\delta)$) that the fitted body reproduces; constant within 0.02 for $\delta = 2$, 5 and 10 %. 300 people per sex.

	Arm	Forearm	Thigh	Calf
Men	0.81	0.82	0.81	0.85
Women	0.50	0.60	0.49	0.66

carry more uncertainty (more estimated fat), so the prior weighs more: about half of a real difference would be drawn. How large real differences are, and whether the fit follows them, is open.

Width of the prior. Scaling the population covariance by 1, 2 or 3 in standard deviation changes every predicted tape by at most 0.2 cm on ANSUR II (limb mean with all tapes: 1.60, 1.53, 1.61 cm for men; 1.53, 1.51, 1.51 cm for women); a factor of 2 is marginally best within this homogeneous population, and the narrower prior removes the drag of the trunk tapes on the men's biceps (1.33 and 1.35 cm with all tapes at factors 1 and 2). We keep 3 as a design choice for training populations; the only evidence is anecdotal (with factor 1, a real thigh 15 % above typical was flagged as an outlier during development). For women the modelled noise accounts for nearly all the observed between-person variance at most sites, so their covariance sits at its floor and some correlations saturate (thigh–hips); skinfold measurements would separate fat from muscle.

Performance. Table 2 reports medians on an NVIDIA RTX 4060 Ti (Chrome, ANGLE/Direct3D 11) and, as a pessimistic bound, with CPU rasterization (SwiftShader, Intel i7-12700F). An idle viewer renders nothing.

Reproducibility. Refactoring the whole code base for release (version 0.2) was verified end to end: the rebuilt models have byte-identical binary chunks, the calibration reproduces bit for bit, all 377 activation names of the previous version resolve to the same targets, and 17 rendering scenarios — poses, split view, automatic views, shapes, tape rings — match pixel for pixel under deterministic software rendering. Every number in this section comes from a script in the repository.

10 Limitations and future work

Model. The model is one male body: female proportions are skeletal transforms and every person shares one skeleton scaled to two heights. Regions thicken as a whole — the biceps cannot grow apart from the triceps — and a tape informs its entire region. The trunk is matched as a ratio over the typical body, so skeletal breadth is not personalized. The isotropic limbs trade volume fidelity per level for round sections, and the quadriceps heads keep the source's proportions. The model shows no fat or skin, and the head lacks facial and masticatory muscles, which BodyParts3D does not contain.

Evaluation. Body composition was a proxy estimated from tapes or BMI, never a reference method. The calibration population is U.S. Army personnel aged 17–58; beyond FFMI 28 (men) and 21 (women) the curves extrapolate, and at the top of the range the quadriceps look bulbous (Fig. 3). Three questions need people, and we release protocols and analysis scripts for each: whether the fit follows *real* left–right differences (both sides of every limb measured on 20–40 adults); whether the prior, and its factor of 3, suit a *trained* population (30 or more lifters with bioimpedance or DXA, analysed for spreads 1–3); and whether coaches agree with the automatic views (3–5 raters choose the best of eight rendered azimuths for 30 exercises, analysed with Fleiss' κ [6] and the algorithm's agreement with the raters compared with theirs among themselves).

Future work. A covariance of the per-site fat residuals, learnt on ANSUR II like the muscle covariance, would let the fit use what the regression exploits; skinfold input would separate fat from muscle directly. Per-muscle morphs, a translucent fat layer from the fitted fat, exercise animation with the existing rig, EMG-derived activation intensities and measurements on mobile GPUs are further steps.

11 Conclusion

body3d shows that an anatomically detailed, personalized muscle viewer fits the constraints of ordinary web pages. Linearity is the thread through the design: the layered hypertrophy splits into exactly additive regional morph targets; probes and a volume polynomial make the tape fit a small, smooth least-squares problem; and a population prior learnt from anthropometric data lets a few tape measures shape the whole body while conserving its muscle mass. The evaluation also measures the price of that consistency: given many tapes, a plain regression predicts a hidden girth better, because the model explains every tape with muscle alone. The code is released under the MIT licence and the models under CC BY 4.0.

Availability. Source, models, documentation (API, architecture, pipeline, design decisions, validation) and the scripts reproducing every number in this paper are in the body3d repository: `dev/validate-ansur.mjs` (Table 1), `dev/asymmetry-ansur.mjs` (Table 4), `dev/muscle-volumes.mjs` (Table 3), `pipeline/check_isotropy.py` and `pipeline/check_fit.py` (geometry), and the study kits `dev/validate-measured.mjs` and `dev/view-study.mjs`.

Acknowledgments. The models derive from BodyParts3D, © The Database Center for Life Science, licensed under CC Attribution 4.0 International. The calibration uses the public ANSUR II data of the U.S. Army. body3d was first developed for the Trace training tracker, whose exercise catalog provided the training vocabulary and the views evaluated here.

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